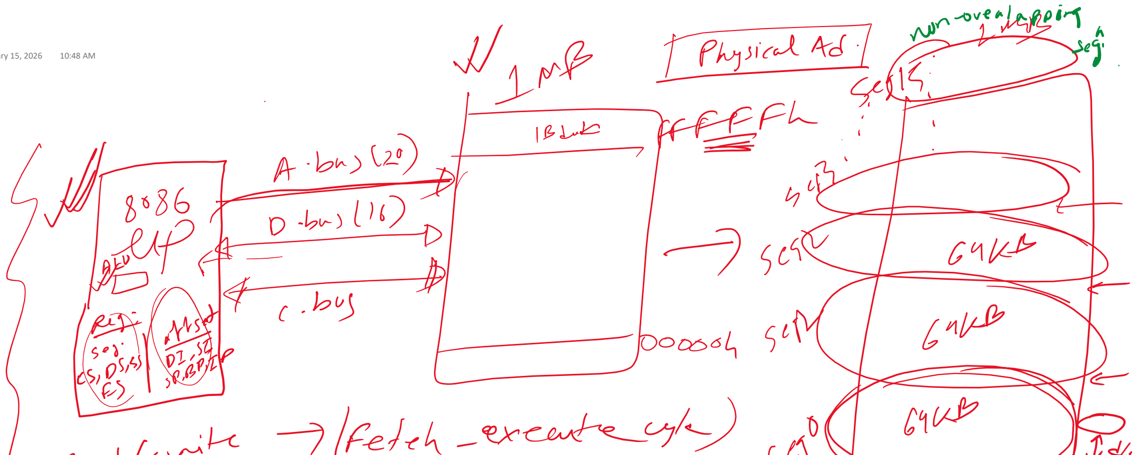
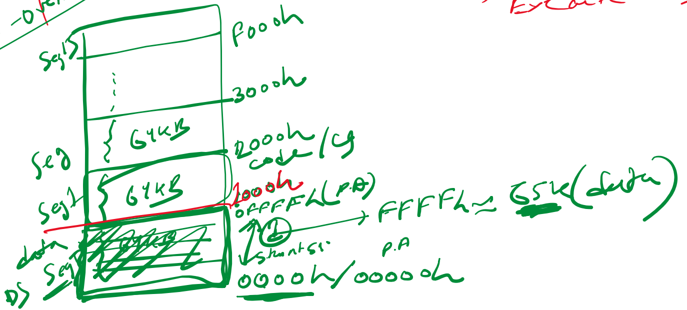

8086
uP
ALU
Reg:
seg:
CS, DS, SS
ES
offset
DI, SI
SP, BP, IP
A.bus (20)
D.bus (16)
c.bus
1MB
1Byte
FFFFFh
00000h
Physical Ad.
non-overlapping
1 64KB
seg.
seg15
seg3
seg2
seg1
seg0
64KB
64KB
64KB
d/ofs
seg

Read/write → (fetch_execute_cycle)

→ Fetch
→ Decode
→ Execute

logical add = seg : offset
4 bit Hex num : 4 bit hex

non-overlapping:

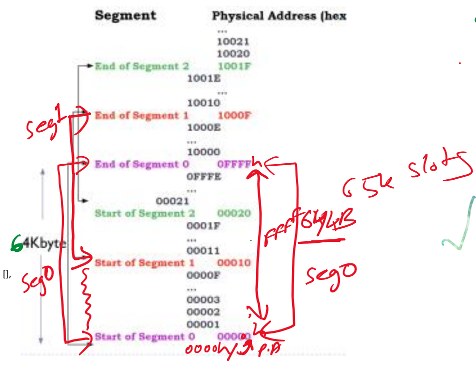

Seg15
F000h
3000h
64KB
2000h / cs
code
Seg
64KB
Seg1
1000h
0FFFFh (P.A)
data
DS Seg0
64KB
①
Short si
0000h / 00000h
P.A

→ FFFFh ≃ 65k (data)

(iii) 0001~~0~~h
~~10h~~
20001h

(ii)

Overlapping segmentation

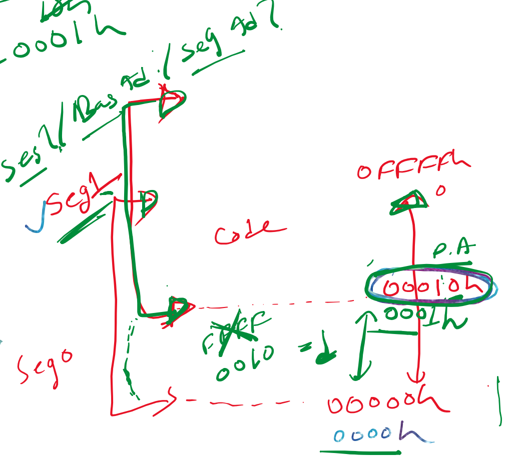


Segment **Physical Address (hex**

...
10021
10020
End of Segment 2 1001F
1001E
...
10010
End of Segment 1 1000F
1000E
...
10000
End of Segment 0 0FFFF
0FFFE
...
00021
Start of Segment 2 00020
0001F
...
00011
Start of Segment 1 00010
0000F
...
00003
00002
00001
Start of Segment 0 00000

64Kbyte

seg1
seg0
h
65k slots
FFFF 64KB
seg0
0000h
P.A


seg1 / Base ad. / seg ad?
seg1
code
seg0
0FFFFh
P.A
00010h
0001h
~~FFFF~~
0010 = d
00000h
0000h

(ii) In Overlapping segmentation, for a physical address there can be more than 1 base address / segment Add. / segment number.

P.A → 00010h is a part of overlapping seg. now find whats the bases add?

(i)
Base add1 = 0000h
Base add2 = 0001h

P.A = seg × 10h + offset

00010
10h
seg1 0001

✓ P.A = $seg_1 \times 10h$ + offset

⇒ 00010 = 0001 × 10h + offset

⇒ 00010 = 00010 + offset ?

⇒ offset = (00010 − 00010)h

= 0000h

seg1 10h
2 0001 0

PA 00010h

Logical add:

	seg	:
①	0001	0000h
②	0000	0010h

2 logical add

② P.A = $seg_2 \times 10h$ + offset (→ 5 bit)

⇒ 00010 = 0000 × 10h + offset

⇒ 00010 = 00000h + offset

⇒ offset = (00010 − 00000h)

= 0010h

✓ (iii) In overlapping segmentation 1 P.A can have multiple logical addresses respect to it.

(iv) In overlapping segmentation, the base addresses will always start from such a P.A which is divisible by 10h / has a 0 at the end. eg →

1030h ✓
00010h ✓
ABCD0h ✓
ABCBh ✗
DC132h ✗

9) ① Yes overlap,

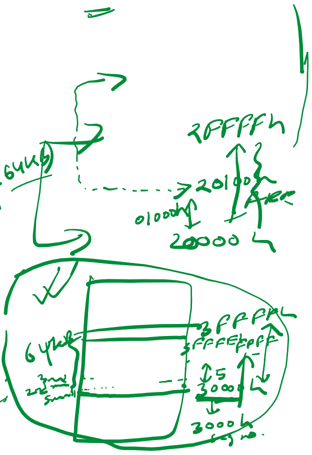
(20100h − 20000h) = 0100h (offset)

FFFFh ✗

over 2 load

≈ 65K / 64KB offset ✗ seg ✗

12. ① A0822h.


(64KB)
3FFFFh
20100h
0100h
20000h
3FFFFh
3FFFEh
2FFFF
5
3000h
64KB
3rd
2nd
small
3000h
seg no.

15. P.A = 32556h → seg: 3000h

5th smallest:

P.A = seg × 10h + offset

= 3000 × 10h + 0004h

= 30004h

5th largest

P.A = Seg × 10h + offset

= 3000 × 10h + FFFBh

= 3FFFBh

Logic Ad:

Logical:

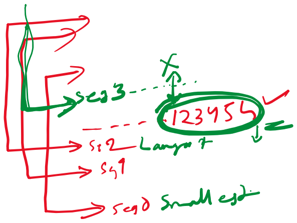
= 5

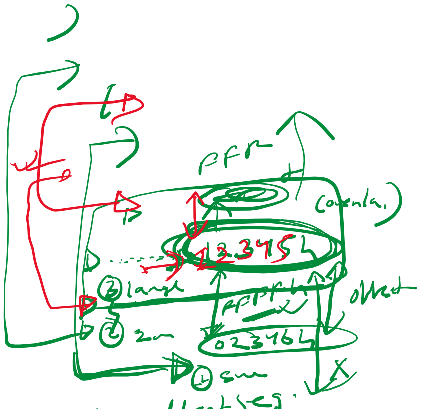
Login A1:
3000 : 0004h

Logical 1:
3000 : FFFBh

Q.4

Smallest sg:


seg 3
12345h
seg 2 Largest
seg 1
seg 0 Smallest


FFR
(overlap)
12345h
③ largest
② 2nd
① 1st
FFFFh
offset
02346h
X

✓smallest seg.

largest seg?

Steps:

largest offset

ⓘ (12345h − FFFFh) = 02346h

This P.A is not divisible by 10h so, we need to roundup/ round down the last digit to 0.

(ii) 02346h → 02350 / 02340h

seg → 0235h / 0234h

(iii) P.A = seg * 10h + offset

⇒ 12345 = 0235 * 10h + offset

⇒ 12345 − 02350 = offset

⇒ offset = FFF5h

P.A = seg * 10h + offset

⇒ 12345 = ~~0234~~ * 10h + offset

⇒ 12345 − 02340 = offset

⇒ ~~offset~~ = ~~10005h~~

5 bit hex

$$02350 + FFF5 = 12345$$

So the valid smallest segment is = 0235h

largest!

ⓘ (12345h − 0000h) = 12345h
~~0h~~

(ii) 12340, 12350

(iii)